# Homework 2

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## 1. Question 1

*1.a.*

Yes, see reasoning in 1b.

*1.b.*

Define the payoff as $X$, price vector as $P$, and price of contingent claim as $p$. Solve the linear equation $Xp = P$ by elimination, we got

$$[X, P] \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0.2 \\ 0 & 1 & 0 & 0.4 \\ 0 & 0 & 1 & 0.2 \end{bmatrix} \tag{1}$$

Thus, we have complete markets in this economy and the price of continguent claim for state of nature 1, 2,3 are $[0.2, 0.4, 0.2]'$

*1.c.*

The price should be

$$[30, 20, 10] \times [0.2, 0.4, 0.2]' = 16 \tag{2}$$

*1.d.*

SDP in this economy is calucalteed by $M(i) = \frac{p(i)}{\pi(i)}$

$$M = [0.8, 0.8, 0.8]'. \tag{3}$$

This economy satisfies no-arbitrage as all the prices of contingent claims are strictly positive and unique. SDF remains the same accross all states, which means that agents in this economy are risk neutral.

*1.e.*

$$P_D = \sum_i \pi(i) M(i) X(i) = \frac{1}{4} \cdot 0.8 \cdot 30 + \frac{1}{2} \cdot 0.8 \cdot 20 + \frac{1}{4} \cdot 0.8 \cdot 10 = 16 \tag{4}$$

Homework
Solution

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## 2. Question 2

*2.a.*

nm

| Parameter | Estimate | Std. error | t-stat |
|---|---|---|---|
| $a$ | 1.0368 | 0.0119 | 86.86 |
| $b_1$ (Lo 10) | −1.5850 | 1.6173 | −0.98 |
| $b_2$ (2-Dec) | 3.5503 | 1.8540 | 1.91 |
| $b_3$ (3-Dec) | 0.7104 | 2.0766 | 0.34 |
| $b_4$ (4-Dec) | −3.1122 | 1.8756 | −1.66 |
| $b_5$ (5-Dec) | 1.1837 | 2.0043 | 0.59 |
| $b_6$ (6-Dec) | 2.9972 | 1.9857 | 1.51 |
| $b_7$ (7-Dec) | −5.2931 | 1.8685 | −2.83 |
| $b_8$ (8-Dec) | 2.4722 | 1.6310 | 1.52 |
| $b_9$ (9-Dec) | 2.8775 | 1.6288 | 1.77 |
| $b_{10}$ (Hi 10) | −1.1771 | 1.0598 | −1.11 |

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 19.

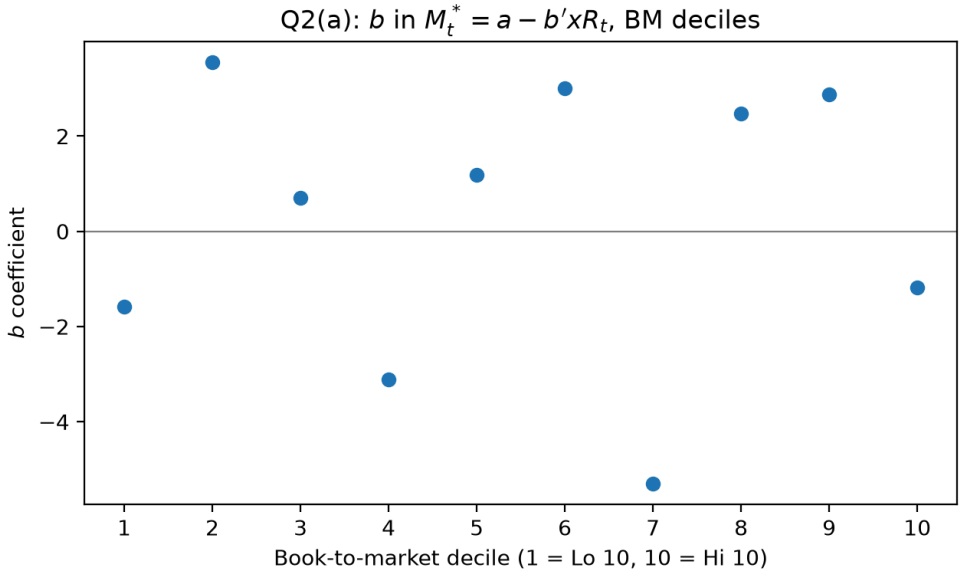



Figure 1: $b$ coefficients of $M_t^* = a - b'xR_t$ by book-to-market decile (two-step GMM, 1927:07–2026:08).

## 3. Question 3

*3.a.*